

MINI-ASSESSMENT

Multi-Step Prompt Workflow – AI Resume Analyzer & Job Match Assistant

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1. Objective

The objective of this workflow is to design a multi-step Generative AI prompt workflow that analyzes a candidate's resume against a job description, identifies matching skills, detects skill gaps, generates improvement recommendations, and evaluates the final result.

The workflow uses prompt chaining, information extraction, classification, skill matching, gap analysis, validation, and iterative refinement.

2. Workflow Overview



3. Detailed Workflow Steps

Step 1 – Extract Candidate Information

Purpose

The first step extracts important information from the candidate's resume and converts unstructured information into structured data.

Prompt

You are an AI Resume Analysis Assistant.

Analyze the provided resume and extract:

1. Education
2. Technical skills
3. Soft skills
4. Projects
5. Work experience
6. Certifications
7. Relevant achievements

Organize the information using clear headings and bullet points.

Do not evaluate the candidate yet.

Do not invent information that is not present in the resume.

Output

A structured candidate profile containing education, skills, projects, experience, certifications, and achievements.

4. Step 2 – Analyze Job Requirements

Purpose

This step analyzes the job description and extracts the requirements needed for the position.

Prompt

You are an AI Job Description Analysis Assistant.

Analyze the provided job description and identify:

1. Required technical skills
2. Preferred technical skills
3. Required experience
4. Educational requirements
5. Key responsibilities
6. Important keywords
7. Required soft skills

Clearly separate mandatory requirements from preferred requirements.

Do not add requirements that are not present in the job description.

Output

A structured job requirement profile.

5. Step 3 – Match Skills and Experience

Purpose

This step compares the candidate's profile with the job requirements.

Prompt

Using the candidate information from Step 1 and the job requirements from Step 2, compare the candidate against the position.

Classify each major requirement as:

- Strong Match
- Partial Match
- No Match

For every classification, provide a concise reason based only on the available resume and job-description information.

Do not invent qualifications, experience, or skills.

Present the result in a structured table:

Requirement	Candidate Evidence	Match Level	Reason
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Output

A detailed skill and requirement matching table.

6. Step 4 – Identify Skill Gaps

Purpose

This step identifies the skills that the candidate needs to improve for the target role.

Prompt

Using the matching analysis from Step 3, identify the candidate's skill gaps.

For each gap, provide:

1. Missing or weak skill
2. Importance to the job
3. Evidence from the job description

4. Suggested learning or improvement area

Do not identify a skill as missing if it is clearly present in the resume.

If there is insufficient information to determine whether a skill is present, mark it as "Needs Verification."

Output

A structured skill-gap analysis.

7. Step 5 – Generate Improvement Suggestions

Purpose

This step creates practical recommendations based on the identified skill gaps.

Prompt

Using the skill-gap analysis from Step 4, generate practical recommendations for improving the candidate's suitability for the role.

Include:

1. Resume improvement suggestions
2. Technical skills to strengthen
3. Project ideas that would demonstrate relevant skills
4. Keywords that can be highlighted when truthful
5. Interview preparation areas

Prioritize recommendations based on their importance to the target position.

Do not recommend false claims, fabricated experience, or skills that the candidate does not possess.

Output

A prioritized career improvement plan.

8. Step 6 – Final Quality Evaluation

Purpose

This step evaluates the complete analysis for accuracy, relevance, and reliability.

Prompt

Evaluate the complete resume and job-match analysis against the original resume and job description.

Score each criterion from 1 to 5:

1. Accuracy
2. Relevance
3. Completeness
4. Evidence-based reasoning
5. Consistency
6. Practical usefulness
7. Hallucination risk
8. Professional presentation

Calculate the overall score out of 40.

Then provide:

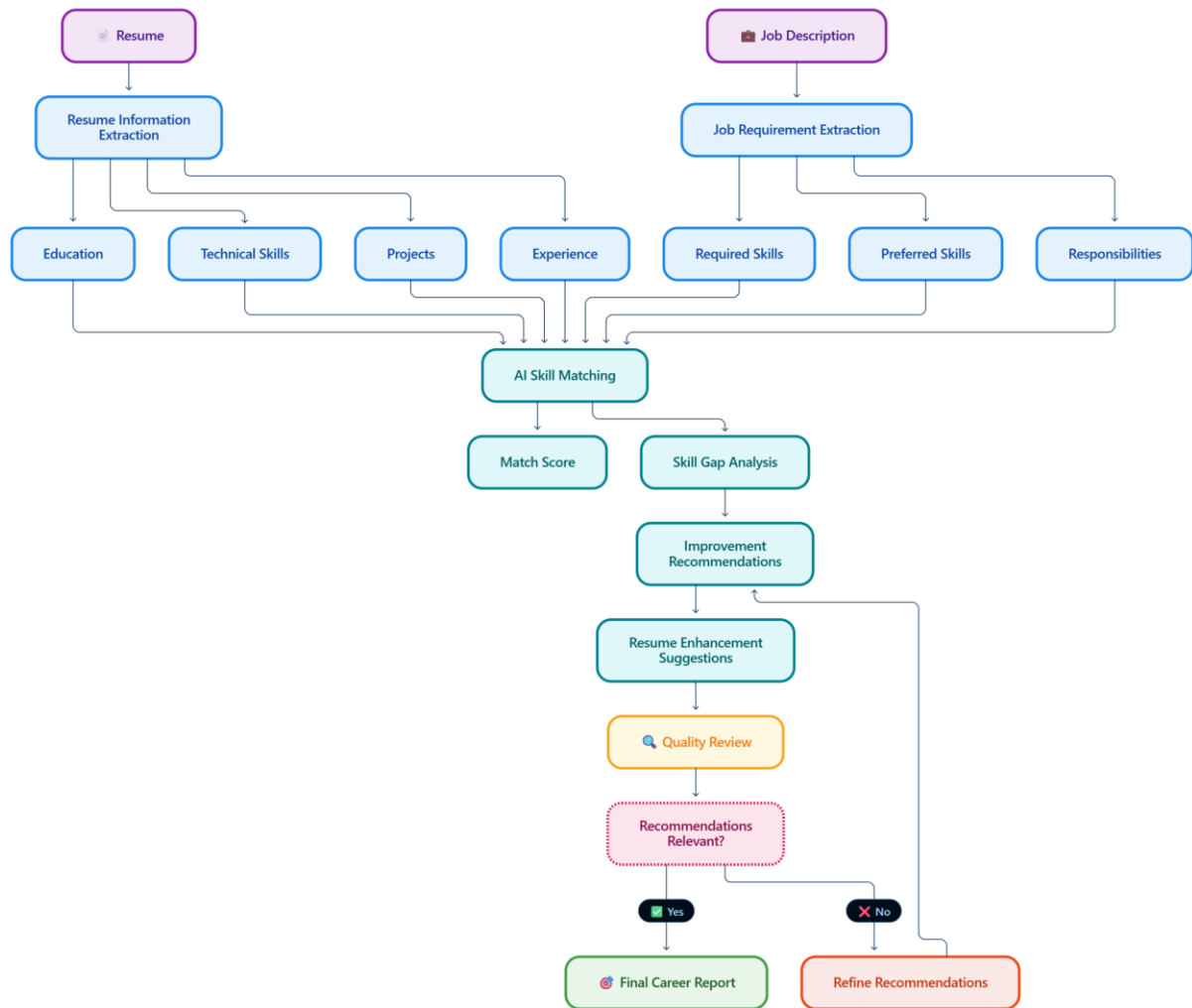
- Overall assessment
- Three major strengths
- Three weaknesses
- Specific improvements required before final submission

Do not invent information or rewrite the resume.

Output

A final quality score with strengths, weaknesses, and improvement recommendations.

9. Detailed AI Workflow



10. Input and Output Flow

Step	Input	Processing	Output
1. Candidate Extraction	Resume	Information extraction	Candidate profile
2. Job Analysis	Job description	Requirement extraction	Job requirement profile
3. Skill Matching	Candidate + requirements	Comparison and classification	Match analysis
4. Gap Analysis	Match analysis	Skill-gap identification	Skill gaps
5. Recommendations	Skill gaps	Recommendation generation	Career improvement plan

Step	Input	Processing	Output
6. Evaluation	Complete analysis	Quality assessment	Score and feedback

11. Example Input

Resume

Software engineering student with knowledge of Python, Java, SQL, React, Git, and machine learning. Completed academic projects involving web development and data analysis.

Job Description

Entry-level Software Engineer requiring Python or Java, SQL, Git, problem-solving skills, and basic cloud knowledge. Candidates should be able to work collaboratively and develop reliable software solutions.

Expected Workflow Result

The AI should identify:

- Python → Strong Match
- Java → Strong Match
- SQL → Strong Match
- Git → Strong Match
- Problem Solving → Needs Verification / Partial Match depending on resume evidence
- Cloud → Skill Gap if not present
- Collaboration → Needs Verification if not demonstrated

The system can then recommend learning basic cloud concepts and adding relevant evidence to the resume if genuinely available.

12. Key Generative AI Concepts Demonstrated

1 Prompt Chaining

The output from each stage is passed as context to subsequent stages.

2 Information Extraction

The AI converts unstructured resume and job-description information into structured data.

3 Classification

Candidate requirements are classified into Strong Match, Partial Match, or No Match.

4 Skill Matching

Candidate capabilities are compared with job requirements based on available evidence.

5 Gap Analysis

The workflow identifies missing or weak skills relevant to the target role.

6 Structured Prompting

Every prompt defines a role, objective, constraints, and expected output.

7 Iterative Refinement

Recommendations can be refined when the quality review identifies problems.

8 AI Evaluation

The final analysis is measured against predefined quality criteria.

13. Advantages of the Workflow

1. Saves time when comparing resumes with job descriptions.
2. Provides a consistent evaluation structure.
3. Identifies matching skills clearly.
4. Highlights important skill gaps.
5. Produces practical career recommendations.
6. Reduces hallucination risk by requiring evidence from the provided information.
7. Supports iterative improvement.
8. Can be reused for different job roles.
9. Helps candidates prepare for interviews.
10. Produces a structured and easy-to-understand career report.

14. Expected Final Result

The workflow produces a personalized career report containing:

- Candidate profile
- Job requirements
- Skill-match analysis
- Match classification
- Skill gaps
- Improvement recommendations

- Interview preparation areas
- Final quality score

The final report is based on the information provided by the user and avoids fabricating qualifications or experience.

15. Conclusion

The AI Resume Analyzer and Job Match Assistant demonstrates how Generative AI can be used as a structured multi-step workflow for career analysis.

The workflow combines information extraction, requirement analysis, skill matching, classification, gap analysis, recommendation generation, quality review, and iterative refinement.

Instead of relying on a single prompt, the workflow decomposes the task into specialized stages. This improves organization, consistency, transparency, and usefulness of the final output.

The validation and refinement loop further improves reliability by ensuring that recommendations remain relevant to the original resume and job description.